UPPSALA UNIVERSITET
Matematiska institutionen
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Prov i matematik
Geometri och Analys II, 1MA188,
2025-05-26

Time: 08:00 - 13:00.

Tools allowed: only pen/pencil and paper.

The exam consists of **8 problems** of **5 points** each for a total of **40 points**. You need 18, 24, 32 points for grade 3, 4, 5, respectively.

Note: if you received Problem 1, 2, or 3 bonus from your dugga result you may skip the corresponding problem but clearly write "5/5 because of dugga".

Good luck and have fun!

1. (5 pts) Let T be the $\mathbb{R}^2 \to \mathbb{R}^2$ linear transformation with the standard matrix

$$[T] = \begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix}.$$

(a) Find the standard matrix of the composition transformation $T \circ T$.

(b) Compute all the eigenvalues and all the eigenvectors of the matrix $[T]$.

(c) Given that T is a projection onto a line in $\mathbb{R}^2$, find the equation of this line.

2. (5 pts) Let $x \in \mathbb{R}$ and

$$A = \begin{pmatrix} x & x & x & x \\ 1 & x & x & x \\ 2 & 2 & x & x \\ 0 & 2 & -1 & x \end{pmatrix}$$

(a) Compute $\det A$.

(b) Determine for which values of x the matrix A is invertible.

3. (a) (2.5 pts) Compute the limit or show that it does not exist

$$\lim_{(x,y)\to(0,0)} \frac{y\sin(x^2+y^2)}{x^2+y^2}$$

(b) (2.5 pts) Compute the limit or prove that it does not exist:

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^2y^2 + x^2z + yz}{x^2+y^2+z^2}$$

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4. Let Y_1 be the surface given by the equation

$$z = x^2 + y^2$$

and Y_2 be the surface given by

$$z = -(x+1)^2 - y^2 + 1.$$

Let γ be the curve of intersection of Y_1 and Y_2.

(a) (1pt) Find the tangent plane to Y_1 at the point $(0,0,0)$.

(b) (1pt) Find the tangent plane to Y_2 at the point $(0,0,0)$.

(c) (1pt) Find the tangent line to γ at the point $(0,0,0)$ (*Hint: use (a) and (b)*).

(d) (1pt) Find the projection of γ onto the xy-plane (find its equation and sketch).

(e) (1pt) Find the projection of γ onto the xz-plane (find its equation and sketch).

5. (5 pts) Solve the partial differential equation

$$\frac{\partial z}{\partial x} + 3x^2 \frac{\partial z}{\partial y} = xy$$

by using the change of variables $u = x^3 - y$, $v = x$.

6. (5pts) Let $f(x,y) = x^2 + 2y^2 + x^2 y$.

(a) Find all the critical points of f.

(b) Classify each of the critical points as a local minimum, local maximum, or a saddle point.

7. (5pts) Let D be the region in $\mathbb{R}^2$ bounded by $y = x - 1$ and $y^2 + 2y - x + 1 = 0$. Sketch D and compute its area.

8. (5pts) Let E be the region in $\mathbb{R}^3$ given by $x^2 + y^2 + 9z^2 \leq 9$, $z \geq 0$. Compute

$$\iiint_E (x + y + z)\, dx\, dy\, dz$$